be a homomorphism defined by

$$\lambda \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = \lambda_1(a)\lambda_2(c)$$

for some pair of homomorphisms $\lambda_1, \lambda_2 : \mathbb{F}_q^\times \to \mathbb{C}^\times$. Define

$$V_{\lambda_1,\lambda_2} = \mathrm{Ind}_B^G \mathbb{C}_\lambda,$$

where $\mathbb{C}_\lambda$ is the 1-dimensional representation of B in which B acts by λ. We have

$$\dim(V_{\lambda_1,\lambda_2}) = \frac{|G|}{|B|} = q+1.$$

Theorem 5.25.2. (1) *$\lambda_1 \neq \lambda_2 \Rightarrow V_{\lambda_1,\lambda_2}$ is irreducible.*

(2) *$\lambda_1 = \lambda_2 = \mu \Rightarrow V_{\lambda_1,\lambda_2} = \mathbb{C}_\mu \oplus W_\mu$, where W_μ is a q-dimensional irreducible representation of G.*

(3) *$W_\mu \cong W_\nu$ if and only if $\mu = \nu$; $V_{\lambda_1,\lambda_2} \cong V_{\lambda_1',\lambda_2'}$ if and only if $\{\lambda_1, \lambda_2\} = \{\lambda_1', \lambda_2'\}$ (in the second case, $\lambda_1 \neq \lambda_2, \lambda_1' \neq \lambda_2'$).*

Proof. From the Frobenius formula, we have

$$\mathrm{Tr}_{V_{\lambda_1,\lambda_2}}(g) = \frac{1}{|B|} \sum_{a\in G,\, aga^{-1}\in B} \lambda(aga^{-1}).$$

If

$$g = \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix},$$

the expression on the right evaluates to

$$\lambda(g)\frac{|G|}{|B|} = \lambda_1(x)\lambda_2(x)\big(q+1\big).$$

If

$$g = \begin{pmatrix} x & 1 \\ 0 & x \end{pmatrix},$$

the expression evaluates to

$$\lambda(g)\cdot 1,$$

since here $aga^{-1} \in B \Rightarrow a \in B$.

If

$$g = \begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix},$$